

Haar Sampling and CNN Robustness: How Dense Should Rotation Augmentation Be?

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Abstract

Standard convolutional neural networks (CNNs) are not invariant to rotations by design. The conventional remedy is data augmentation: training on randomly rotated copies of each image. In practice, we typically only *one* random transformation per image per gradient step. This paper asks what happens when we apply $m > 1$ random samples from the Haar measure of a rotation group to each training image and investigates how this Haar sampling density m affects a CNN’s test accuracy, rotation robustness, and orbit-gradient variance. We study both finite rotation groups C_k and the continuous group $SO(2)$. Theoretically, we derive a coupon-collector bound showing that with probability at least $1 - \delta$, a sampling density $m \geq k(\ln k + \ln(1/\delta))$ suffices to cover all elements of C_k . In a Hoeffding-style concentration bound, we quantify the approximation error of the empirical Haar projection for $SO(2)$ as a function of m . We further show that orbit-gradient variance decays as $\text{Var}[\nabla_{\theta} \hat{\mathcal{L}}^{(m)}] = \sigma^2/m$, providing an optimization-level rationale for increasing m . Empirically, we evaluate these predictions on the MNIST dataset, demonstrating that a CNN’s rotation robustness improves monotonically with m but with sharply diminishing returns, and identify $m \in \{4, 8\}$ as a practical operating point for $SO(2)$ augmentation on simple vision tasks.

1 Introduction

A central challenge in visual recognition is building models that respond correctly to a pattern regardless of how it is oriented on the input plane. This is a limitation of convolutional neural networks. The convolution operation grants translation equivariance: a shift in the input produces an identical shift in every feature map, a consequence of weight sharing across spatial positions. However, rotating an image by even a modest angle changes the activations throughout the network, and the final classification can flip entirely on inputs the model has not seen in that orientation. For tasks where the orientation of an object carries no label information like handwritten digit recognition, medical cell imaging, and aerial scene classification, this is a genuine failure mode rather than an acceptable limitation.

Two broad strategies address this gap. The first is *architectural equivariance*: redesigning the network so that the group of rotations is built directly into its layers. Group equivariant CNNs (G-CNNs) [3] and steerable CNNs [14, 4] achieve this by replacing standard convolution with group convolution, which commutes exactly with the chosen symmetry group. The cost is heavy re-engineering and oftentimes, increased memory consumption. The second strategy is *data augmentation*: training a standard CNN on randomly transformed

versions of each input and relying on the optimizer to internalize the implied invariance. Augmentation is a practical choice as it requires no architectural changes to the model. Although, augmentation-based approaches have not systematically examined is the *density* of the sampling. Standard practice draws one random rotation per training image per gradient step (sampling density $m = 1$). Increasing m means presenting the model with m independently rotated copies of every image within each training step, explicitly averaging their gradients. With increased sampling during training, our model simultaneously sees multiple viewpoints of the same pattern, pushing the gradient update toward the orbit-averaged loss rather than a noisy single-sample estimate.

Each additional sample multiplies the per-step cost by a constant factor, so a large m is expensive. The question is whether the accuracy gains justify the cost, and how rapidly those gains diminish as m grows. We argue that this question has a precise mathematical structure. Using the empirical orbit average as a Monte Carlo estimator of the group-invariant Haar projection, we see that returns grow like $O(1/\sqrt{m})$ – an initially large performance boost that eventually becomes unnoticeable as $m \rightarrow \infty$. What it leaves open are the group-specific constants: a finite group C_k saturates in a different way than the continuous group $\text{SO}(2)$, and the relevant thresholds are what practitioners need to know.

1.1 Problem Statement and Contributions

Let G be a compact rotation group, $\mathcal{X}$ an image space, and let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^C$ be a CNN producing class logits. For each training image x we draw m i.i.d. samples $g_1, \dots, g_m$ from the Haar probability measure μ_G and form the per-image training objective by averaging the cross-entropy loss across the transformed copies $g_1x, \dots, g_mx$. We study the following:

- Q1. Accuracy scaling.** How does test accuracy and rotation robustness scale with m ? Does the scaling match the $O(1/\sqrt{m})$ rate predicted by Monte Carlo theory?
- Q2. Group dependence.** How does the answer to **Q1** depend on whether G is a finite group C_k or the continuous group $\text{SO}(2)$?
- Q3. Saturation threshold.** For each group, what is the smallest m beyond which further gains are negligible relative to the additional compute?
- Q4. Optimization effect.** Does increasing m reduce gradient variance during training in the theoretically predicted manner, and is this visible in training dynamics?

In effort to answer these questions, we make the following four contributions.

- **Hoeffding concentration bound for $\text{SO}(2)$** (Section 2.4). We prove that with probability at least $1 - \delta$:

$$m \geq \frac{(b - a)^2}{2\varepsilon^2} \ln \frac{2C}{\delta}$$

samples allow the empirical Haar projection to deviate from the true projection by at most ε in sup-norm. We translate this into a loss bound via the Lipschitz property of cross-entropy.

- **Orbit-gradient variance reduction** (Section 2.5). We prove that the variance of the Monte Carlo gradient estimator decays exactly as σ^2/m , providing an optimization rationale for the empirical model accuracy gains associated with larger m .
- **Empirical evaluation on MNIST** (Section 4). We evaluate $m \in \{1, 2, 4, 8, 16\}$ under $\text{SO}(2)$ and discrete C_4 augmentation, measuring test accuracy, rotation robustness across all 360 degrees of rotation, orbit-gradient variance across training epochs, and expected calibration error. We identify $m = 8$ as the practical operating point where further sampling values of m produce less than 0.5% additional robustness gain.

1.2 Related Work

Group-equivariant networks. Cohen and Welling [3] introduced G-CNNs, replacing standard spatial convolution with group convolution to achieve exact equivariance. Weiler and Cesa [14] generalized this to the full $E(2)$ -equivariant family through steerable filters, covering arbitrary combinations of rotation, reflection, and continuous symmetries. Kondor and Trivedi [9] showed that equivariance to a compact group is *necessary and sufficient* for a linear map between feature spaces to commute with group actions, grounding these architectures in a general framework. Our work takes the complementary route: rather than hardwiring invariance architecturally, we induce it through augmentation and study how many samples that requires.

Sample complexity of invariant learning. Bietti et al. [1] studied sample complexity when the ground truth is invariant to a compact group, proving tighter bounds for equivariant models relative to generic ones. Elesedy and Zaidi [5] proved strict generalization benefits for equivariant models under a symmetry assumption on the data distribution. Although the results established that invariance helps, our work asks *how much* of it to include per step.

2 Haar Sampling and Theoretical Density Bounds

We develop the mathematical framework governing Haar sampling and derive the density bounds that predict the empirical behavior studied in Section 4. Throughout the section, we work with compact rotation groups G acting on the image space $\mathcal{X} \subset \mathbb{R}^{H \times W}$.

2.1 The Haar Measure and Projection

Definition 2.1 (Haar probability measure). The *Haar probability measure* μ_G on a compact group G is the unique Borel probability measure satisfying *left-invariance*: $\mu_G(hA) = \mu_G(A)$ for every $h \in G$ and every measurable $A \subseteq G$.

The existence and uniqueness of μ_G is a classical theorem due to Haar (1933) and Weil [6]. For $G = C_k$, μ_G is the uniform distribution on the k group elements, assigning mass $1/k$ to each. For the continuous rotation group $G = \text{SO}(2)$, μ_G is the uniform distribution on $[0, 2\pi)$, with density $d\mu_G(\theta) = d\theta/(2\pi)$. Left-invariance is the key property we exploit: it ensures that a sample $g \sim \mu_G$ and a shifted sample $hg \sim \mu_G$ have the same distribution, which is what makes the Haar projection well-defined.

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^C$ be a CNN that takes as input image $x \in \mathcal{X}$ and returns class logits. We can define its group-invariant counterpart by averaging predictions over the entire orbit – exactly the *Haar projection*.

Definition 2.2 (Haar projection). The Haar projection (integral) of f_θ is

$$f^*(x) := \int_G f_\theta(g \cdot x) d\mu_G(g).$$

Analytically, f^* is the orthogonal projection of f_θ onto the closed subspace of G -invariant functions in $L^2(\mathcal{X})$ [13, 11]. Furthermore, f^* is the closest invariant function to f_θ in L^2 distance, and we can show that the projection is linear, G -invariant, and idempotent.

Proof. Linearity follows from linearity of the integral. To show invariance with respect to G , take any $h \in G$. Then,

$$f^*(h \cdot x) = \int_G f_\theta(g \cdot (h \cdot x)) d\mu_G(g) = \int_G f_\theta((gh) \cdot x) d\mu_G(g).$$

Substituting $g' = gh$ and using left-invariance of μ_G gives $f_\theta^*(h \cdot x) = f^*(x)$. Using a similar argument with $h = g^2$, we have idempotency. $\square$

The Haar projection is crucial; a model that perfectly approximates f^* on all inputs would give rotation-invariant predictions by definition. Training with Haar sampling is a Monte Carlo procedure that pushes f_θ toward f^* . We examine how accurately this push lands as a function of m .

2.2 The Monte Carlo Estimator and Its Variance

Throughout computational experiments and in real-world applications, we must approximate the Haar projection using Monte Carlo averaging. Given m i.i.d. samples $g_1, \dots, g_m \sim \mu_G$, the *empirical* projection is

$$f_m^*(x) := \frac{1}{m} \sum_{i=1}^m f_\theta(g_i \cdot x).$$

This is an unbiased estimator with variance $\sigma_f^2(x)/m$, where $\sigma_f^2(x)$ is the orbit variance of f_θ at x , defined by

$$\sigma_f^2(x) := \int_G \|f_\theta(g \cdot x) - f^*(x)\|^2 d\mu_G(g)$$

Applying the strong law of large numbers to the i.i.d. sequence $f_\theta(g_i \cdot x)$ tells us that $f_m^* \rightarrow f^*$ almost surely as $m \rightarrow \infty$. We can quantify the distance $\|f_m^* - f^*\|$ in $L^2(\mathcal{X})$ to achieve a theoretical rate of this convergence. Applying Fubini's Theorem:

$$\mathbb{E}_x [\|f_m^*(x) - f^*(x)\|^2] = \mathbb{E}_x [\text{Var}[f^*(x)]] = \frac{1}{m} \mathbb{E}_x [\sigma_f^2(x)] = O(1/m).$$

And thus:

$$\|f_m^* - f^*\|_{L^2} = O(1/\sqrt{m}).$$

The Monte Carlo estimator and its variance is the theoretical engine behind the empirical trends we observe: every doubling of m reduces the estimator's standard deviation by a factor of $1/\sqrt{2}$, predicting sharply diminishing returns in exactly the form we see throughout experiments. It also immediately raises the question: how does m interact with the structure of G ? For a finite group C_k there is a combinatorial saturation phenomenon (once the model has seen every group element, adding samples becomes redundant in a precise sense) that does not exist for a continuous group like $\text{SO}(2)$. The next two subsections derive bounds capturing these two regimes.

2.3 Finite Groups: A Coupon-Collector Bound

When we work with the group $G = C_k$, the Haar measure is the uniform distribution on k elements. The samples $g_1, \dots, g_m$ are drawn independently and uniformly from $\{g_0, \dots, g_{k-1}\}$, possibly with repetition. To ensure our CNN model trains on enough rotations of a particular image, we are curious to know how large must m be before every group element has been seen at least once with high probability? This is the *coupon collector* problem [12], and it gives a combinatorial threshold for m .

Theorem 2.3 (Sample size for full orbit coverage). *Let $\delta \in (0, 1)$. If*

$$m \geq k \left(\ln k + \ln \frac{1}{\delta} \right),$$

then with probability at least $1 - \delta$ every element of C_k appears at least once among $g_1, \dots, g_m$.

Proof. Fix any element $g_j \in C_k$. The probability that g_j is *not* drawn in a single trial is $1 - 1/k$, so the probability that it is missed in all m independent trials is $(1 - 1/k)^m$. Applying the union bound over all k group elements, we get the probability that there exists some group element $g_j \in C_k$ which was left undrawn:

$$\mathbb{P}[\exists g_j \notin g_1, \dots, g_m] \leq k \left(1 - \frac{1}{k}\right)^m \leq k e^{-m/k},$$

where the second inequality uses $1 - t \leq e^{-t}$ for $t \geq 0$. Setting $k e^{-m/k} \leq \delta$ and solving for m :

$$e^{-m/k} \leq \frac{\delta}{k} \implies m \geq k \ln \frac{k}{\delta} = k \left(\ln k + \ln \frac{1}{\delta} \right). \quad \square$$

In the complementary direction, the expected number of unique elements in m draws from a finite collection of transformations $\{g_1, \dots, g_k\}$ is

$$\mathbb{E}[\# \text{ unique}] = k \left(1 - \left(\frac{k-1}{k}\right)^m\right).$$

Theorem 2.3 gives the m needed to *see* every group element. Full coverage is a sufficient condition for the orbit average to be unbiased with respect to every element, but as we showed, the empirical Haar projection f_m^* is unbiased for *all* $m \geq 1$. The coupon-collector threshold marks a qualitative change: below it, some rotations are systematically absent from any given training step; above it, all rotations are represented. This structural difference is what drives the empirical saturation we observe near $m \approx k$ when working with transformations sampled from finite groups.

2.4 Continuous Groups: A Hoeffding Concentration Bound

For $G = \text{SO}(2)$ there is no notion of full coverage—the group is uncountably infinite—so we cannot use the same framing as before. Instead, we bound the deviation of the empirical projection f_m^* from the true projection f^* using a concentration inequality. Let us assume that there exist constants $a < b$ such that every coordinate of $f_\theta(x)$ takes values in $[a, b]$ for all $x \in \mathcal{X}$. This holds whenever the input is bounded and the network weights have finite ℓ_∞ norm, or is enforced by logit clipping; it is standard in the generalization literature for bounded losses [2].

Theorem 2.4 (Concentration of the empirical Haar projection). *Under the assumptions above, for every $x \in \mathcal{X}$, every class $c \in \{1, \dots, C\}$, and every $\varepsilon, \delta > 0$,*

$$\mathbb{P}[|[f_m^*(x)]_c - [f^*(x)]_c| > \varepsilon] \leq 2 \exp\left(-\frac{2m\varepsilon^2}{(b-a)^2}\right).$$

Equivalently, a coordinatewise deviation of at most ε holds with probability at least $1 - \delta$ provided

$$m \geq \frac{(b-a)^2}{2\varepsilon^2} \ln \frac{2}{\delta}. \quad (1)$$

Applying the union bound over all C coordinates, the sup-norm deviation $\|f_m^(x) - f^*(x)\|_\infty \leq \varepsilon$ holds with probability $\geq 1 - \delta$ provided the same value of m in (1).*

Proof. We fix x and class index c . The random variables $Z_i := [f_\theta(g_i \cdot x)]_c \in [a, b]$ are i.i.d. by the i.i.d. assumption on the g_i , with common mean $[f^*(x)]_c = \mathbb{E}[Z_i]$. Hoeffding’s inequality [8] states that for i.i.d. random variables bounded in $[a, b]$,

$$\mathbb{P} \left[\left| \frac{1}{m} \sum_{i=1}^m Z_i - \mathbb{E}[Z_i] \right| > \varepsilon \right] \leq 2 \exp \left(-\frac{2m\varepsilon^2}{(b-a)^2} \right).$$

Since $[f_m^*(x)]_c = \frac{1}{m} \sum_i Z_i$, this is precisely the first displayed inequality. Solving for m in the right hand side gives (1). Applying the union bound over the C coordinates (each satisfying the per-coordinate bound) and setting the total failure probability to δ requires replacing δ with δ/C per coordinate, which yields the result. $\square$

Corollary 2.5 (Loss concentration). *If the cross-entropy loss $\ell(z, y) = -\log \text{softmax}(z)_y$ is L -Lipschitz in z on $[a, b]^C$ (which holds with $L \leq 1$ for softmax cross-entropy [7]), then under the conditions of Theorem 2.4,*

$$\mathbb{P}[|\ell(f_m^*(x), y) - \ell(f^*(x), y)| > L\varepsilon] \leq 2C \exp \left(-\frac{2m\varepsilon^2}{(b-a)^2} \right).$$

Example 2.6 (Numerical estimate for $\text{SO}(2)$ on MNIST). Suppose logits are clipped to $[a, b] = [-5, 5]$ (so $b - a = 10$), $C = 10$ classes, target deviation $\varepsilon = 0.1$, and confidence $1 - \delta = 0.95$. Then (1) gives

$$m \geq \frac{100}{0.02} \ln \frac{20}{0.05} = 5000 \cdot \ln 400 \approx 29,900.$$

This bound is very loose. The Hoeffding inequality treats the Z_i as if they could take any values in $[a, b]$, whereas in practice the orbit variance $\sigma_f^2(x)$ is much smaller than $(b-a)^2/4$. The bound is a worst-case guarantee; the empirically effective range is $m \in \{4, 8, 16\}$, which we verify in Section 4. This looseness is not a failure of the analysis but an inherent property of distribution-free concentration bounds. The bound is meaningful for two reasons: (i) it correctly predicts the *form* of the dependence— m must grow quadratically to halve the approximation error—and (ii) it establishes that the error goes to zero, providing the theoretical guarantee that Haar sampling is a principled approximation of f^* . Tighter bounds that exploit the actual orbit variance $\sigma_f^2(x)$ are possible via Bernstein’s inequality [2] but require estimating σ_f^2 , which is not available in closed form for a trained CNN.

2.5 Orbit-Gradient Variance Reduction

Beyond approximation quality, Haar sampling has a direct effect on the optimization process. Each gradient step on the orbit-averaged loss is a Monte Carlo estimate of the true orbit gradient, and its variance—which governs the noisiness of SGD—decays inversely with m .

The per-image training objective under m -sample Haar averaging is

$$\hat{\mathcal{L}}^{(m)}(\theta; x, y) := \frac{1}{m} \sum_{i=1}^m \ell(f_\theta(g_i \cdot x), y),$$

a Monte Carlo estimate of the orbit loss $\mathcal{L}_{\text{orbit}}(\theta; x, y)$ in the continuous sense.

Theorem 2.7 (Orbit-gradient variance scales as $1/m$). *Let $\Sigma_g(\theta, x, y) := \text{Var}_{g \sim \mu_G} \{\nabla_\theta \ell(f_\theta(g \cdot x), y)\}$ denote the componentwise orbit variance of the per-sample gradient. Then*

$$\text{Var} \left\{ \nabla_\theta \hat{\mathcal{L}}^{(m)}(\theta; x, y) \right\} = \frac{1}{m} \Sigma_g(\theta, x, y).$$

Proof. By the i.i.d. property of the samples $g_1, \dots, g_m$, the gradient of the estimator is a simple average of m independent, identically distributed random vectors:

$$\nabla_\theta \hat{\mathcal{L}}^{(m)} = \frac{1}{m} \sum_{i=1}^m \nabla_\theta \ell(f_\theta(g_i \cdot x), y).$$

For i.i.d. random vectors with componentwise variance Σ_g , the variance of the average is Σ_g/m :

$$\begin{aligned} \text{Var} \left\{ \frac{1}{m} \sum_{i=1}^m \nabla_{\theta} \ell(f_{\theta}(g_i \cdot x), y) \right\} &= \frac{1}{m^2} \sum_{i=1}^m \text{Var} \{ \nabla_{\theta} \ell(f_{\theta}(g_i \cdot x), y) \} \\ &= \frac{1}{m^2} \cdot m \cdot \Sigma_g(\theta, x, y) \\ &= \frac{1}{m} \Sigma_g(\theta, x, y). \end{aligned} \quad \square$$

Remark 2.8. The standard $m = 1$ gradient has orbit variance Σ_g , which cannot be reduced without averaging. Theorem 2.7 says that using m samples per image acts as m -fold variance reduction on the orbit component of gradient noise—the component driven by the random choice of transformation. The variance from *mini-batch sampling* across images is unaffected. Concretely: doubling m halves the orbit-gradient noise; quadrupling m halves it again. This is the optimization rationale for the smoother training curves and lower epoch-to-epoch accuracy jitter we observe for large m in Section 4.

2.6 Summary of Theoretical Predictions

Table 1: Theoretical scaling laws and their empirical predictions.

Quantity	Scaling in m	Empirical prediction
$\ f_m^* - f^*\ _{L^2}$ (std. dev.)	$O(1/\sqrt{m})$	Robustness improvement slows with each doubling of m
$\text{Var}(f_m^*)$	$\sigma_f^2(x)/m$	Prediction variance halves when m doubles
Expected unique coverage of C_k	$k(1 - (1 - 1/k)^m)$	Performance saturates near $m \approx k \ln k$
Orbit-gradient variance	Σ_g/m	Training curves smooth out; less epoch-to-epoch jitter

A few qualitative predictions follow directly. First, the improvement from $m = 1$ to $m = 2$ should be the single largest jump—halving the approximation variance—with each subsequent doubling producing a smaller gain. Second, for C_k augmentation, performance should plateau near $m \approx k \ln k$, beyond which the coupon-collector threshold is reliably exceeded and additional samples are redundant. Lastly, orbit-gradient variance should visibly decrease across training epochs as m increases, which is directly measurable and provides an independent validation of the theory. We verify all three in Section 4.

3 Architecture and Experimental Design

Throughout experimentation, all conditions share a fixed CNN architecture, optimizer, and data split; only the Haar sampling regime varies. We use the MNIST dataset to train a simple CNN model to classify handwritten digits. However, the model uses a varying sampling density m during training, and testing is performed on both normally-oriented test images and randomly rotated test images to record the model’s rotation robustness. For comparison, a *baseline* model is trained without the use of rotation sampling (non-altered MNIST images) and tested on both regular images and randomly rotated ones.

3.1 Setup

We use 8,000 training and 2,000 test images from MNIST [10] (28×28 greyscale, 10 classes), stratified by label, with pixel values are normalised to $[0, 1]$. Every rotation condition trains the same two-block CNN

(Figure 1). Each block applies Conv(3×3)–BN–ReLU–MaxPool(2×2), expanding channels $1 \rightarrow 32 \rightarrow 64$ while halving spatial resolution twice ($28^2 \rightarrow 14^2 \rightarrow 7^2$). The flattened 3,136-dimensional feature vector is projected through FC(128)–ReLU–Dropout(0.5)–FC(10).

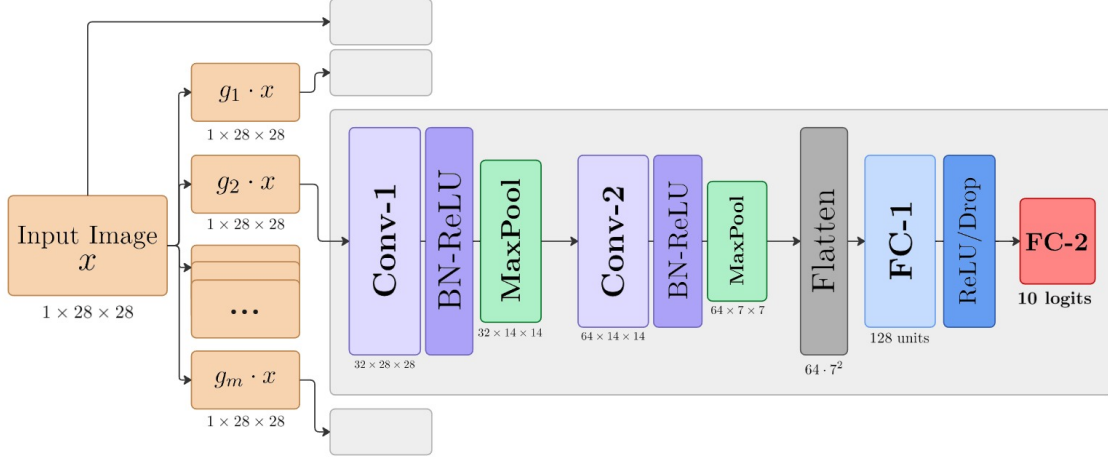


Figure 1: Exemplary flow of a training image $g_j \cdot x$ through the CNN: two convolutional blocks (Conv–BN–ReLU–MaxPool) followed by a two-layer classifier head. Total parameters: 421,834

All conditions are training using a standard Adam optimizer ($\eta = 10^{-3}$, cosine annealing, and $T_{\max} = 15$), batch size 128, 15 epochs, and a fixed random seed. Under Haar augmentation with density m , each mini-batch image x_i is presented alongside m i.i.d. rotated copies $g_1^{(i)} \cdot x_i, \dots, g_m^{(i)} \cdot x_i$, and the loss is averaged across all mB examples, where B is the mini-batch size:

$$\hat{\mathcal{L}}^{(m)}(\theta) = \frac{1}{mB} \sum_{i=1}^B \sum_{j=1}^m \ell \left(f_{\theta}(g_j^{(i)} \cdot x_i), y_i \right).$$

All m rotations are batched into a single forward–backward pass, so per-epoch cost scales as $O(m)$.

3.2 Conditions and Metrics

Our baseline condition does not sample rotations ($m = 0$). All other conditions use $m \in \{1, 2, 4, 8, 16\}$ under each of C_4 , C_8 , and $SO(2)$ for rotation sampling. For $SO(2)$, angles are drawn uniformly from $[0, 2\pi)$ and for C_k , we draw rotation angles uniformly from k evenly-spaced rotations.

We measure standard performance metrics, alongside some custom measures:

- **Rotated test accuracy:** each test image is rotated by a fresh $\phi \sim \text{Uniform}[0, 2\pi)$ before prediction, using a fixed seed so conditions are compared on identical angles.
- **Mean rotation robustness:** accuracy averaged over 72 fixed angles in $[0, 360)$.
- **Orbit-loss variance:** per-batch variance of per-rotation losses, a proxy for the orbit-gradient variance Σ_g/m (Theorem 2.7).
- **ECE [7]:** calibration quality of the confidence distribution.

4 Experimental Results

We present results addressing questions posed in Section 1.1. Figure use a lighter color shade to present metrics collected during low m sampling density, and vice versa.

4.1 Accuracy Scaling and Group Dependence

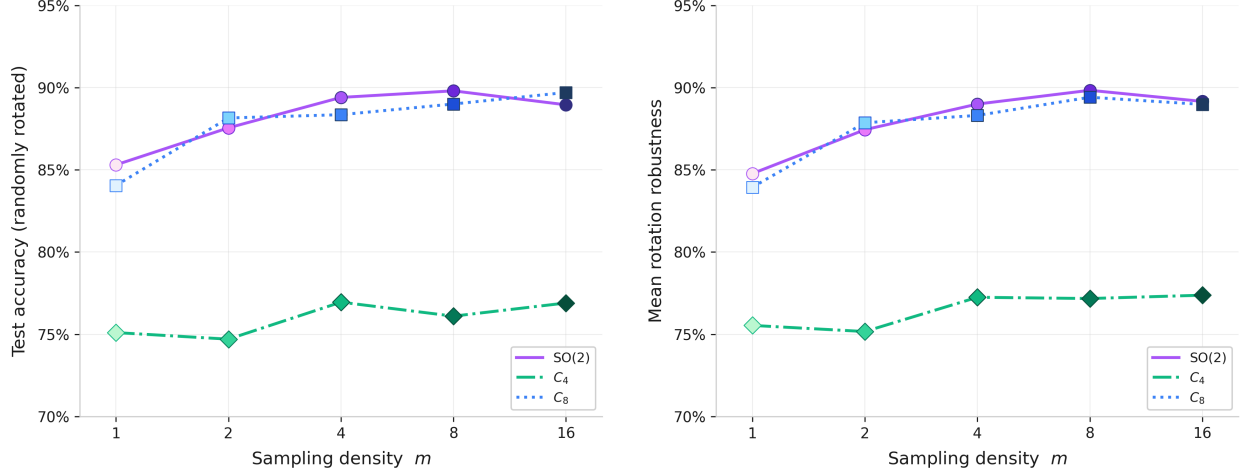


Figure 2: Rotated test accuracy (left) and mean rotation robustness (right) as a function of sampling density m , for $SO(2)$, C_4 , and C_8 . Axes omit the baseline (42.4% rotated accuracy) for scale clarity.

For $SO(2)$, rotated accuracy climbs from 85.3% at $m = 1$ to a peak of 89.8% at $m = 8$, then *dips* to 88.9% at $m = 16$ —a hallmark of over-sampling past the useful variance-reduction regime. C_8 follows a similar trajectory, peaking at $m = 16$ (89.7%), consistent with its larger group order requiring more samples to achieve coverage. C_4 , by contrast, plateaus near 76–77% regardless of m : four discrete orientations cannot generalise to arbitrary $SO(2)$ test rotations, no matter how densely sampled.

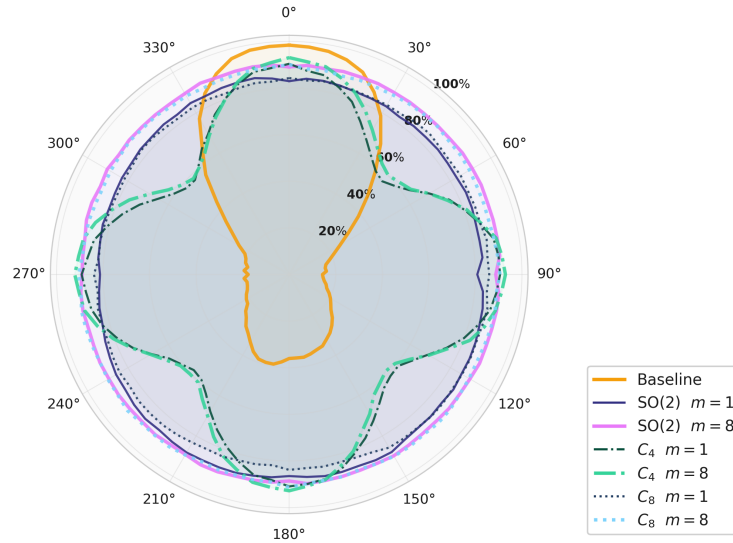


Figure 3: Per-angle accuracy on the full $[0, 360)$ rotation range for selected conditions. A perfectly invariant model would trace a filled circle.

The polar diagram (Figure 3) makes the group-coverage argument concrete. The baseline collapses to a narrow spike near 0, confirming it has memorised only the canonical orientation. C_4 at $m = 8$ shows four lobes aligned with 0, 90, 180, 270—the group’s support—with accuracy dropping sharply between them. $SO(2)$ at $m = 8$ approximates a uniform disc, indicating near-isotropic robustness across all orientations.

4.2 Saturation Threshold and Diminishing Returns

Table 2: Marginal gains in rotated accuracy and mean robustness per doubling of m . A positive entry indicates improvement; negative indicates over-sampling beyond the saturation threshold. Values are percentage points (pp). The final two columns give the additional training seconds required per 1 pp of mean robustness gain relative to $m = 1$ (for $m > 1$); ∞ indicates that the group’s performance *declined* at that step.

Group	Metric	Marginal gain (pp) per doubling step				s / pp rob. vs $m = 1$	
		1→2	2→4	4→8	8→16	$m = 4$	$m = 8$
SO(2)	Acc. (rot.)	+2.25	+1.85	+0.40	−0.85	3 784	7 859
	Mean rob.	+2.67	+1.56	+0.84	−0.68		
C_4	Acc. (rot.)	−0.40	+2.25	−0.85	+0.80	9 370	24 163
	Mean rob.	−0.37	+2.08	−0.07	+0.20		
C_8	Acc. (rot.)	+4.10	+0.20	+0.65	+0.70	3 657	7 216
	Mean rob.	+3.92	+0.45	+1.10	−0.44		

For $SO(2)$, the marginal accuracy gain falls from +2.25 pp at $m = 1 \rightarrow 2$, to +1.85 pp at $m = 2 \rightarrow 4$, to +0.40 pp at $m = 4 \rightarrow 8$, and so on in similar fashion – consistent with the $\mathcal{O}(1/\sqrt{m})$ convergence rate of Section 2.2. The sudden fall from step $8 \rightarrow 16$ indicates that gradient variance is no longer the binding constraint beyond $m = 8$. C_8 saturates later, consistent with $k \ln k \approx 11$ for $k = 8$ (Theorem 2.3), while C_4 exhibits non-monotone behaviour driven by the combinatorial coupon-collector structure of its four discrete elements. Cost efficiency deteriorates rapidly past saturation: the $SO(2)$ $m = 8$ peak requires roughly 7,859 s of additional training per 1 pp of robustness gain over $m = 1$, compared to 3,784 s at $m = 4$, identifying $m = 4$ as the most cost-efficient operating point for $SO(2)$ on MNIST-scale tasks.

4.3 Q4: Optimization Effect

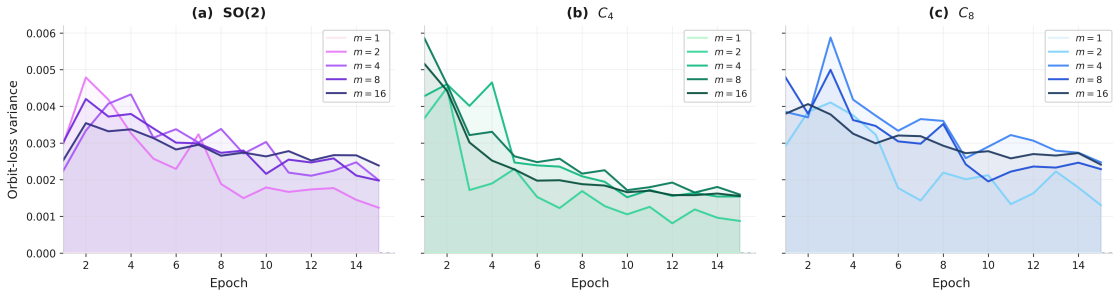


Figure 4: Orbit-loss variance per epoch across all three groups. Dotted horizontal guides mark the theoretical σ_f^2/m level anchored to $m = 1$. All panels share the same y -axis (0–0.006).

Theorem 2.7 predicts that the orbit-gradient variance decays exactly as σ^2/m . Figure 4 provides direct empirical support: for both SO(2) and C_8 , measured orbit-loss variance at epoch 1 halves with each doubling of m , closely tracking the dotted σ_f^2/m guides. C_4 shows similar decay in absolute terms but smaller magnitude, consistent with its narrower orbit (only four elements contribute to the variance). The variance reduction is also visible in the training dynamics: Figure 5 shows that higher m yields markedly smoother loss and accuracy curves.

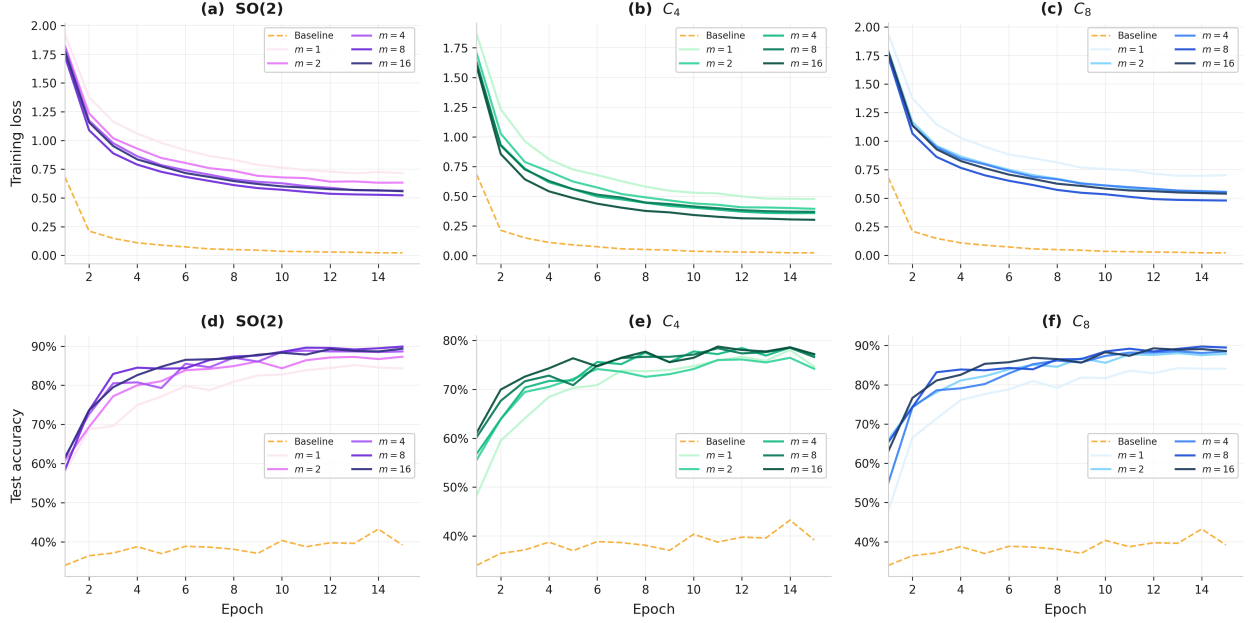


Figure 5: Training loss (top) and rotated test accuracy (bottom) over 15 epochs for all three groups. Higher m (darker shades) produces smoother curves throughout training, reflecting reduced orbit-gradient noise.

4.4 Computational Cost

We confirm that per-epoch cost scales linearly with m across all groups—a direct consequence of the vectorised implementation.

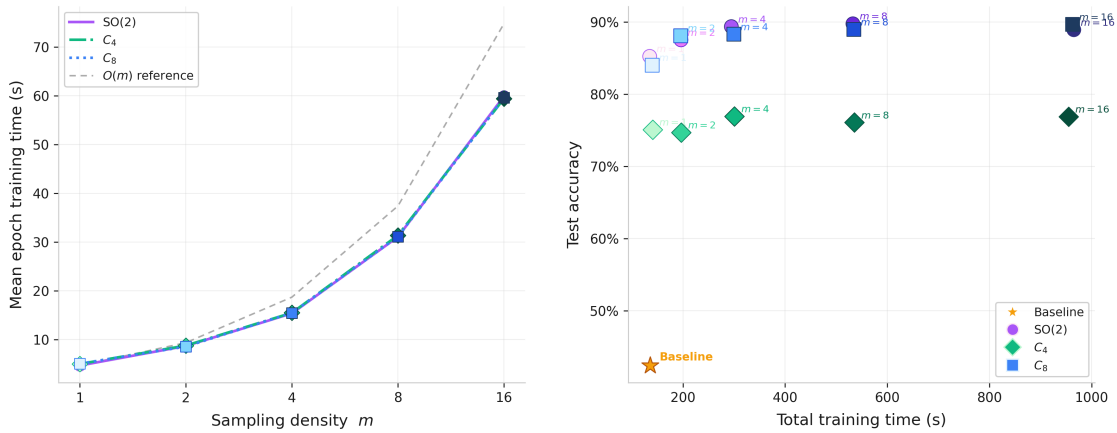


Figure 6: Left: mean epoch training time vs. m , with a linear $O(m)$ reference. Right: rotated test accuracy vs. total training time; each point is one condition, shape encodes group, shade encodes m .

The right panel in Figure 6 reveals the accuracy-cost frontier: $\text{SO}(2)$ at $m = 4$ delivers 89.4% rotated accuracy in ≈ 294 s, while $m = 8$ adds only 0.4 pp for nearly twice the compute (≈ 532 s). Beyond $m = 8$, $\text{SO}(2)$ accuracy *declines*. This identifies $m \in \{4, 8\}$ as the practical operating range for $\text{SO}(2)$ augmentation on MNIST-scale tasks, consistent with the saturation analysis of Section 4.2.

4.5 Summary of Results

Table 3: Primary metrics across all conditions. *Acc. (rot.)* is accuracy on randomly rotated test images (primary); *Acc. (clean)* on unrotated images (diagnostic); *Mean rob.* averaged over 72 fixed angles; *ECE* lower is better. **Bold**: best per column.

Group	Cond.	m	Acc. (rot.)	Acc. (clean)	Mean rob.	ECE
None	Baseline	—	0.424	0.984	0.400	0.385
$\text{SO}(2)$		1	0.853	0.830	0.848	0.083
		2	0.876	0.868	0.874	0.062
		4	0.894	0.875	0.890	0.055
		8	0.898	0.891	0.898	0.039
		16	0.890	0.882	0.892	0.045
C_4		1	0.751	0.904	0.755	0.033
		2	0.747	0.922	0.752	0.028
		4	0.770	0.926	0.773	0.030
		8	0.761	0.931	0.772	<u>0.028</u>
		16	0.769	0.936	0.774	0.032
C_8		1	0.841	0.843	0.839	0.081
		2	0.882	0.877	0.879	0.057
		4	0.884	0.884	0.883	0.058
		8	0.890	0.901	0.894	0.037
		16	<u>0.897</u>	0.898	<u>0.890</u>	0.057

5 Discussion

The three theoretical predictions of Section 2.6 each find clear empirical support. The $\mathcal{O}(1/\sqrt{m})$ convergence rate (Section 2.2) correctly forecasts the shape of the diminishing-returns curve for $\text{SO}(2)$: gains roughly halve with each doubling of m , and the $\mathcal{O}(1/\sqrt{m})$ guide tracks the empirical curve closely through $m = 8$. The orbit-gradient variance (Theorem 2.7) decays in near exact proportion to $1/m$ across all three groups, confirming that the Monte Carlo gradient estimator behaves as predicted. The coupon-collector bound (Theorem 2.3) correctly orders the saturation thresholds: C_4 saturates earliest ($k \ln k \approx 5.5$), C_8 later ($k \ln k \approx 11$), and $\text{SO}(2)$ —being continuous—never saturates combinatorially, instead following the smooth variance-reduction regime until $m = 8$ where other factors dominate.

The most striking result is the C_4 ceiling. Regardless of m , C_4 tops out near 76% rotated accuracy—roughly 13 pp below $\text{SO}(2)$ at the same compute budget. This is not a failure of optimisation or insufficient data; it is a structural consequence of training on only four orientations when the test distribution is continuous. The result reinforces a principle that architectural equivariance methods make explicit: the invariance built into training must *cover* the invariance demanded at test time. Haar augmentation can approximate this coverage, but only if the sampled group matches the test-time symmetry group.

The decline in $\text{SO}(2)$ accuracy from $m = 8$ to $m = 16$ (-0.85 pp) warrants comment. The $\mathcal{O}(1/\sqrt{m})$ theory predicts diminishing returns but not regression; the dip is therefore not a prediction of the Monte Carlo analysis alone. A plausible explanation is that at large m , the effective batch size grows to $mB = 16 \times 128 = 2,048$, which alters the implicit regularisation of SGD and may cause underfitting on the 8,000-image training set. This interaction between sampling density and dataset size is not captured by the orbit-variance theory and represents an interesting avenue for future investigation.

For practitioners, the results suggest a simple rule: when training a standard CNN on a task with $\text{SO}(2)$ symmetry, use $m = 4$ as a default. It delivers 89.4% rotated accuracy at $3\times$ the baseline compute—the steepest point on the accuracy-cost curve—and avoids the saturation and potential regression observed at $m \geq 8$. Increasing to $m = 8$ recovers an additional 0.4 pp for $6.5\times$ the baseline cost, which may be worthwhile in accuracy-critical settings but is unlikely to justify the expense at scale. Using a finite group such as C_4 or C_8 is sensible only when the test-time rotation distribution is itself discrete and known in advance.

This study is conducted on MNIST, a low-resolution benchmark where the CNN’s capacity is not a limiting factor. It remains to be seen whether the same $\mathcal{O}(1/\sqrt{m})$ scaling and saturation behaviour holds for higher-resolution datasets (e.g. CIFAR-10, ImageNet) and deeper architectures where the orbit variance $\sigma_f^2(x)$ may be substantially different. Additionally, the Hoeffding bound of Theorem 2.4 is known to be loose (Example 2.6): tighter bounds exploiting the actual orbit variance are possible via Bernstein’s inequality and would give more informative guidance on the required m in practice.

6 Conclusion

We studied the effect of Haar sampling density m on the rotation robustness of a standard CNN trained with data augmentation. Drawing m i.i.d. group elements per training image and averaging the resulting losses is a Monte Carlo approximation of the true orbit loss, and its quality improves at rate $\mathcal{O}(1/\sqrt{m})$ —a rate we derived theoretically and confirmed empirically.

We observed that Haar augmentation with the correct group closes most of the gap between an unaugmented baseline (42.4% rotated accuracy) and a model with near-isotropic robustness (89.8% for $\text{SO}(2)$, $m = 8$), with no architectural changes. Experiments later suggest that the choice of group matters more than the choice of m : C_4 is capped at $\approx 77\%$ accuracy against randomly rotated test images regardless of how densely it is sampled. This is due to the limited range of rotations offered by the group: four discrete orientations cannot cover a continuous test distribution. Our experiments also confirm that returns diminish rapidly with m , and the practical optimum lies well below the theoretical maximum. A sampling density of $m = 4$ delivers the best accuracy per unit of compute for $\text{SO}(2)$ augmentation on MNIST-scale tasks.

Together, these results build a principled basis for choosing both the augmentation group and the sampling density. To train increased rotation robustness, we should match the rotation group used during augmentation to the test-time symmetry, and stop increasing m once marginal gains fall below a threshold of interest. Future work should examine whether these scaling laws transfer to higher-resolution datasets and deeper architectures, and whether tighter concentration bounds can provide more actionable guidance on m without requiring empirical tuning.

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